Supplementary material

S1. Concepts, stems and the word forms they match (1 Nov 1884–1888, both papers)

Stems are matched as substrings of surface forms; following Schoots (2023b) we inspected forms for false positives (our first *law* stem also matched *inteto*, speech, and was dropped) but did not hand-filter them, so run-together forms (e.g. *rulumenteningomsebenzi*) and missed variants remain. All stems, forms and glosses await review by a first-language isiXhosa speaker; e.g. *rulumente* is glossed 'government' here but 'parliament' by Schoots et al. (2023a, p. 62). Political concepts were fixed before testing; religion was suggested by the classifier and is exploratory.

Concept	Type	Stems	Most frequent matched forms (count; number of distinct forms)
parliament	political	palamente	palamente (1754), ipalamente (47), palamenteni (22), epalamente (14), bapalamente (1), nepalamente (1), yepalamente (1); 11 forms
vote	political	voti	ivoti (301), voti (217), evotini (136), abavoti (98), ngevoti (55), levoti (22), nevoti (22); 115 forms
government	political	rulumente, rulumeni	rulumente (1882), rulumeni (1336), rulumenteni (7), urulumente (5), urulumeni (3), rulumenini (2), serulumente (1); 14 forms
rule	political	mbuso	zombuso (169), umbuso (76), mbuso (60), yombuso (42), kombuso (37), lombuso (28), kulombuso (23); 43 forms
law	political	mteto	umteto (724), lomteto (475), mteto (406), ngomteto (167), ngumteto (94), komteto (76), nomteto (76); 82 forms
rights	political	lungelo	amalungelo (315), ilungelo (288), lungelo (113), nelungelo (54), namalungelo (50), lilungelo (39), elilungelo (37); 115 forms
country	political	lizwe	lizwe (646), ilizwe (292), kwelilizwe (201), elilizwe (101), elizwe (96), belilizwe (76), yelilizwe (52); 78 forms
land	political	mhlaba	umhlaba (836), mhlaba (208), lomhlaba (156), nomhlaba (105), komhlaba (99), somhlaba (95), yomhlaba (94); 135 forms
freedom	political	kululek	inkululeko (27), kukululekile (24), ukululekile (13), ngenkululeko (10), abakululekileyo (8), nkululeko (8), nenkululeko (7); 92 forms
humanity	political	buntu	ubuntu (45), yobuntu (14), nobuntu (8), sobuntu (8), buntu (7), lobuntu (7), zobuntu (6); 29 forms
time	neutral	xesha	ixesha (787), ngexesha (364), xesha (298), amaxesha (243), lixesha (191), ngamaxesha (149), zamaxesha (114); 207 forms
house	neutral	ndlu	indlu (719), izindlu (328), kwindlu (276), ndlu (234), yendlu (227), ngendlu (133), nendlu (130); 193 forms
rain	neutral	mvula	imvula (267), mvula (31), yemvula (23), ezimvula (17), lemvula (16), yimvula (14), zimvula (10); 45 forms
cattle	neutral	nkomo	inkomo (555), zenkomo (258), nkomo (193), yenkomo (93), nenkomo (75), ngenkomo (35), lenkomo (33); 79 forms
religion	exploratory	tixo, kristu, krestu	tixo (684), kristu (207), krestu (118), makristu (7), kristukazi (7), makrestu (6), mkrestu (5); 32 forms

S2. Secondary framing tests across settings

Percentile of the observed shift among 2000 frequency-matched word sets, and one-sided p (uncorrected). Jaccard = shift in a word's ten nearest neighbours (neighbour overlap; Gonen et al., 2020). P+V = *palamente* + *voti*; P+V+R adds *rulumente* (uncertain meaning). Directional Δ: Isigidimi minus Imvo, religious minus political similarity (in z-score units), political concepts, two-sided p against matched sets; the last columns give the control *house*. Three settings survive Bonferroni over five ($p < 0.01$): window 1, window 2, and window 2 with forms joined. Joining partly disjunctive forms (*uku ba*) leaves the global profile gap unchanged. Matched sets are re-drawn for each test; without *land* the pooled set is smaller, so lower power may contribute to the higher p.

Setting	P+V profile	P+V Jaccard	P+V+R profile	P+V+R Jaccard	Δ	Δ p	p w/o land	house Δ	house p
window 1 (tuned)	92 (p=0.083)	96 (p=0.039)	92 (p=0.077)	99 (p=0.007)	+0.68	<0.001	0.010	+1.08	<0.001
window 1, forms joined	99 (p=0.010)	95 (p=0.053)	100 (p=0.001)	100 (p=0.002)	+0.44	0.016	0.088	+1.15	<0.001
window 2	86 (p=0.139)	94 (p=0.061)	90 (p=0.096)	99 (p=0.010)	+0.76	<0.001	0.045	+1.34	<0.001
window 2, forms joined	91 (p=0.094)	98 (p=0.023)	99 (p=0.011)	100 (p=0.002)	+0.58	0.009	0.064	+1.51	<0.001
window 5	62 (p=0.382)	98 (p=0.016)	81 (p=0.189)	100 (p=0.002)	+0.66	0.015	0.256	+0.84	0.061

S3. Era drift: does a concept's rate move across eras within one paper?

Per-million-token rates and ratios (later/earlier, or Imvo/Isigidimi) with issue-bootstrap 95% CIs.

Concept	Comparison	First	Second	Ratio (95% CI)
parliament	Isigidimi: later / earlier era	214	394	1.84 (0.96–4.98)
parliament	Imvo: later / earlier era	1320	758	0.57 (0.50–0.66)
parliament	Imvo 1889-94 / Isigidimi 1876-83 (held-out)	214	758	3.54 (1.98–9.58)
government	Isigidimi: later / earlier era	2197	1095	0.50 (0.39–0.64)
government	Imvo: later / earlier era	2234	1649	0.74 (0.67–0.81)
government	Imvo 1889-94 / Isigidimi 1876-83 (held-out)	2197	1649	0.75 (0.65–0.88)
vote	Isigidimi: later / earlier era	72	614	8.55 (3.51–18.94)
vote	Imvo: later / earlier era	799	921	1.15 (0.88–1.51)
vote	Imvo 1889-94 / Isigidimi 1876-83 (held-out)	72	921	12.82 (7.85–25.92)
religion	Isigidimi: later / earlier era	2541	1678	0.66 (0.50–0.86)
religion	Imvo: later / earlier era	445	612	1.38 (1.08–1.76)
religion	Imvo 1889-94 / Isigidimi 1876-83 (held-out)	2541	612	0.24 (0.20–0.30)

S4. Anchors for the directional test

Anchors are the 15 top-PMI neighbours of each seed pair in held-out text (Isigidimi 1876-83 + Imvo 1889-94), frozen before testing. Religious (15 words; seeds *tixo*, *kristu*): *sikelela*, *sindisa*, *ukumkanikazi*, *inkosazana*, *buka*, *oyingcwele*, *lilizwi*, *msindisi*, *nelizwi*, *kwelizwi*, *wanga*, *lika*, *yesu*, *izono*, *yehova*. Political (15 words; seeds *palamente*, *voti*): *emqulwini*, *wonyulo*, *lonyulo*, *elugqatsweni*, *eziqingata*, *lokunyula*, *kwesase*, *lokuvota*, *endlu*, *kwamagama*, *ukumela*, *izayo*, *ngalomteto*, *nawupina*, *kwabase*. Seed words cannot themselves be tested (Fig. S1). Political anchors include *endlu* ('in the house'), which might have favoured the neutral control *house*; re-running without *endlu* leaves both the political lean and *house*'s lean in place (outputs/framing_matched_tests_noendlu.csv), and *ngalomteto*, which contains the *law* stem. Glosses are unverified: two religious anchors (*ukumkanikazi*, *inkosazana*) may reflect royal rather than religious register, and several anchors on both sides (*lika*, *buka*; *nawupina*, *kwabase*, *izayo*) may be grammatical rather than topical, consistent with narrow-window PMI capturing grammar. All await speaker review.

S5. Readers' letters versus the rest of each paper

Letters are detected by src/12_letters.py: a letter opens by addressing the editor (vocative *mhleli* or *mcokeli*, not *u/ku/ngu mhleli*, which refer to him) or by asking for insertion (*ndifakele*, *kaundifakele*), and ends at the signature (a run of very short lines). A hand check of 30 random detections found 28 genuine letters; the errors were usifake umkala (horse advert: 'put a bit on it'); no mhleli wepepa lama bhulu ('and the editor of the Boer paper'). Recall was not measured. Imvo is restricted to the months in which Isigidimi has text; CIs from 4000 issue resamples; rates are not tested against matched sets. Hits = concept uses.

Part	Concept	Isig. tokens	Isig. hits	Imvo tokens	Imvo hits	Imvo/Isig.	95% CI
letters	parliament	18,344	2	70,475	48	6.25	2.09–16.92
letters	religion	18,344	57	70,475	77	0.35	0.19–0.69
letters	vote	18,344	4	70,475	58	3.77	1.04–18.93
letters	government	18,344	13	70,475	161	3.22	1.67–9.62
rest of paper	parliament	268,319	111	936,998	1280	3.30	2.53–4.51
rest of paper	religion	268,319	424	936,998	370	0.25	0.17–0.37
rest of paper	vote	268,319	172	936,998	821	1.37	0.79–2.83
rest of paper	government	268,319	301	936,998	1906	1.81	1.47–2.28

S6. Which sense of *umhlaba*?

For every use of the land stem (*mhlaba*) we look at the 10 tokens either side. Religious cues: *izulu*, *lizulu*, *nezulu*, and words containing *zulwini*, *tixo*, *kristu*, *yesu*, *msindisi*, *ngcwele* (bare *zulu* is excluded: *kwa zulu* is Zululand). Property/political cues: words containing *taitile*, *mteto*, *rulumente*, *palamente*, *mantyi*, *fama*, *voti*, *nyulo*, *rafu*, *lokishi*. Most uses have neither cue. Example lines for a speaker to check are in outputs/land_kwic.md.

Paper	Cue	Uses	Share	95% CI
Imvo	religious	1408	0.9%	0.4–1.4%
Imvo	property/political	1408	26.8%	23.8–29.7%
Isigidimi	religious	353	3.4%	1.7–5.3%
Isigidimi	property/political	353	12.7%	8.5–17.7%

S7. Same editor, different control; same control, different editor

Cells (editors after Schoots, 2023b, p. 71): Jabavu edits Isigidimi (Jan 1881–Oct 1884); Jabavu edits Imvo (Nov 1884–1888); Gqoba edits Isigidimi (Nov 1884–1888); Makiwane edits Isigidimi (1876–80; Stewart until 1876). Ratios of rates per million tokens; 95% CIs from 4000 issue resamples; p: two-sided, against 2000 matched sets of random words (src/15_editor_control.py). Every contrast is also a contrast in time; *rule* has only 9 uses in Jabavu's Isigidimi, so its large ratios are not interpreted.

Concept	Jabavu: Imvo / Isigidimi			Isigidimi: Gqoba / Jabavu			Isigidimi: 1881–84 / 1876–80		
	ratio	95% CI	p	ratio	95% CI	p	ratio	95% CI	p
parliament	1.69	1.23–2.54	0.088	0.50	0.33–0.79	0.043	49.86	14.41–405.79	$<$0.001
vote	3.07	1.98–5.23	0.017	2.36	1.07–4.61	0.034	11.63	4.32–98.99	0.004
government	1.21	0.99–1.49	0.387	0.59	0.44–0.77	0.043	0.78	0.61–1.01	0.289
rule	19.47	9.46–77.40	0.001	8.28	3.57–31.56	0.009	0.38	0.09–1.32	0.228
law	1.36	1.08–1.72	0.235	0.80	0.53–1.17	0.376	1.31	0.98–1.76	0.314
rights	0.93	0.76–1.16	0.808	0.82	0.58–1.13	0.480	2.91	2.05–4.12	0.006
country	1.08	0.92–1.29	0.773	1.13	0.90–1.41	0.596	0.74	0.59–0.92	0.243
land	1.10	0.91–1.34	0.689	0.95	0.70–1.28	0.853	0.94	0.76–1.16	0.815
freedom	0.68	0.51–0.92	0.443	0.95	0.64–1.38	0.899	1.52	1.01–2.42	0.408
humanity	0.97	0.59–1.90	0.959	1.88	0.98–3.88	0.227	0.97	0.45–1.96	0.967
religion	0.37	0.27–0.50	0.025	1.38	1.03–1.83	0.168	0.35	0.26–0.47	$<$0.001

S8. Hyper-parameter search for the embeddings

fastText skip-gram, tuned on text outside the test window (Isigidimi 1876–83, Imvo 1889–94) so that tuning never sees the compared years. Criterion: seed stability, the mean top-10 nearest-neighbour overlap between two runs that differ only in random seed (Antoniak & Mimno, 2018); unstable embeddings would turn noise into apparent shifts. 34 settings over dimension (50, 100, 200, 300, 400), window (1, 2, 3, 5, 8) and epochs (10, 20, 30), searched in rounds that extended the grid past each optimum. Chosen: dimension 400, window 1, 20 epochs. Stability is highest at the narrowest window, which favours grammar over topic; the paper therefore also reports windows 2 and 5 (Table S2). Best setting per window:

Window	Dimension	Epochs	Isigidimi 1876–83	Imvo 1889–94	Mean stability
1	400	20	0.844	0.814	0.829
2	300	20	0.823	0.798	0.810
3	300	20	0.810	0.799	0.805
5	400	20	0.798	0.796	0.797
8	200	10	0.758	0.755	0.757

At the chosen setting, word2vec reaches a stability of 0.64 against fastText's 0.83, so fastText is used throughout.

S9. Tokenizer cost

Each condition is a random sample of about 100,000 words. Modern text is the isiXhosa section of MasakhaNEWS (Adelani et al., 2023). The old-spelling condition is the same modern sample with the aspiration marks of modern orthography removed by fixed rules (*th*→*t*, *kh*→*k*, *ph*→*p*, *rh*→*r*, *ch*→*c*, *qh*→*q*, *xh*→*x*; src/05_fertility.py), so only spelling changes and the words stay the same: it is compared per word. Historical text differs in word division, so it is compared per 100 characters. The politics/business sample checks genre within modern news: its cost is close to all modern news, while both 1880s papers cost more, so the historical penalty is not simply a news-topic effect, though 1880s and modern newspapers still differ in genre. 95% CIs from bootstrap resampling are within $\pm$0.01 subwords per word.

Condition	Words	XLM-R		Serengeti		mBERT	
		per word	per 100 ch.	per word	per 100 ch.	per word	per 100 ch.
Modern news	100,003	2.613	35.28	1.867	25.21	3.040	41.05
Modern news, old spelling	100,003	2.555	35.56	1.909	26.57	2.937	40.88
Modern news, politics/business only	88,389	2.673	35.21	1.888	24.88	3.116	41.06
Isigidimi 1884–88	100,013	2.259	36.71	1.743	28.33	2.569	41.75
Imvo 1884–88 (matched draw)	100,006	2.240	36.92	1.738	28.65	2.533	41.75

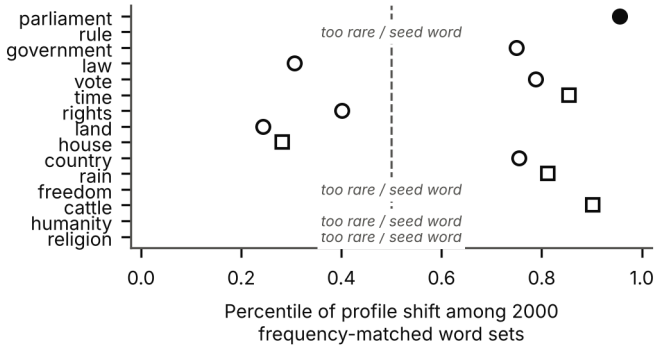


Figure S1: Profile-shift percentile of each concept among matched sets (markers as in Fig. 1, left); 'too rare' = too few occurrences, 'seed word' = anchor source, untestable.

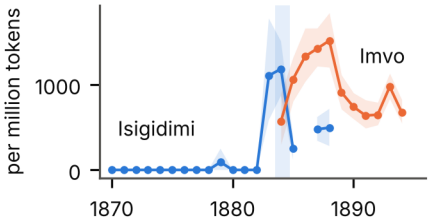


Figure S2: Parliamentary vocabulary per million tokens; it enters Isigidimi in 1883 (light blue: twice-monthly issues, Schoots et al., 2023a, p. 59).